



Coaxial laser-induced breakdown spectroscopy aligned with the welding laser for in-process assessment of Cu–steel laser welding

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ABSTRACT

In this study, laser-induced breakdown spectroscopy (LIBS) was employed using a coaxial optical configuration aligned with a high-power continuous-wave (CW) welding laser to enable real-time acquisition of plasma emission signals during Cu–steel dissimilar metal laser welding. Welding was performed while systematically varying laser power and scanning speed, and the Cu/Fe emission intensity ratio extracted from the coaxially collected LIBS signals was proposed as a criterion for welding quality assessment. The proposed Cu/Fe-based criterion was validated through mechanical joint strength evaluation, demonstrating that LIBS enables in-process, real-time assessment of welding quality. These results indicate that the proposed coaxial, single-laser LIBS approach can be readily integrated into existing laser welding systems without additional excitation sources or complex instrumentation, while enabling accurate and repeatable probing of the weld zone through coaxial signal collection.

1. Introduction

Laser welding has been widely adopted in advanced industrial applications such as semiconductors, secondary batteries, electric vehicles, and aerospace, due to its high precision, low heat input, and suitability for automated manufacturing [1,2]. In particular, the highly localized energy input enables minimization of the heat-affected zone and thermal distortion, while allowing reliable joining of dissimilar and difficult-to-weld materials, thereby providing superior joint quality and process reliability compared to conventional arc and resistance welding methods. With the introduction of high-power laser sources and high-speed beam scanning systems, laser welding processes have achieved significantly enhanced productivity; however, these advancements are also accompanied by increasingly complex melt pool dynamics and process instabilities [3], which necessitate reliable real-time process monitoring.

Laser-Induced Breakdown Spectroscopy (LIBS) is an elemental analysis technique that enables real-time acquisition of material composition by analyzing characteristic atomic and ionic emission lines from laser-induced plasma formed when a high-power laser irradiates a sample surface. LIBS allows standoff detection and operation under ambient conditions without sample preparation, making it a promising

tool for in situ and on-site monitoring in industrial environments [4,5].

In high-speed and high-power laser welding environments, conventional post-process inspection or low-bandwidth sensing approaches are often insufficient for the timely detection of process instabilities, further emphasizing the need for real-time spectroscopic monitoring techniques.

LIBS has been extensively investigated for welding quality monitoring in a variety of welding processes. In non-laser welding processes, such as tungsten inert gas (TIG) and arc welding, Taparli et al. demonstrated that in situ LIBS can be used to monitor the chemical composition of stainless-steel welds during TIG welding by analyzing elemental emission lines originating from the welding plasma [6,7]. Quackatz et al. also investigated in situ LIBS during TIG welding of duplex stainless steels, showing that both qualitative and quantitative compositional information can be obtained using calibration-based and calibration-free LIBS methods [8,9]. Valdiande et al. further demonstrated that plasma spectroscopy combined with LIBS enables online defect detection and process monitoring in arc-welding applications, highlighting the potential of spectroscopic techniques for real-time weld quality assessment in industrial environments [10,11]. These studies clearly established the feasibility of LIBS-based in situ monitoring in welding processes where plasma is inherently generated and continuously

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accessible.

In the context of laser welding, LIBS-based monitoring has also been investigated. Lednev et al. demonstrated online and in situ monitoring of laser welding by employing a separate nanosecond Nd:YAG laser to generate LIBS plasma, while collecting emission signals at an off-axis observation angle [12,13]. These studies showed that LIBS spectral features can be correlated with weld quality and process stability during laser welding of superalloys. However, the requirement for an additional LIBS excitation laser increases system complexity and poses challenges for integration into industrial laser welding systems.

Alternatively, several studies have explored spectroscopic monitoring approaches that utilize plasma generated directly by the welding laser itself, without employing a separate LIBS laser. Sibillano et al. analyzed optical emission from laser-induced plasma during aluminum alloy welding and demonstrated that correlation analysis of spectral signals can be used for real-time defect detection [14]. Ancona et al. demonstrated that spectroscopic analysis of plasma plume optical emission, including electron temperature evaluation, can be used for real-time monitoring of laser welding processes [15]. Palanco et al. investigated the emission characteristics of plasma generated during continuous-wave (CW) laser welding of aluminum alloys and discussed the influence of spectroscopic acquisition conditions on weld quality diagnostics [16]. More recently, Kang et al. employed optical emission spectroscopy combined with neural networks to predict weld cross-sectional geometry during laser dissimilar welding, demonstrating the potential of data-driven approaches for online weld monitoring [17].

In addition, previous studies have also investigated the relationship between spectroscopic signals and welding performance, including correlations with mechanical properties, such as tensile strength, shear load, or fracture behavior during laser welding processes [18]. While these approaches demonstrate that spectroscopic features can be correlated with weld quality, the interpretation of such relationships is often dependent on specific process conditions or system configurations. In this context, the present study aims to provide a more physically interpretable spectroscopic indicator, based on Cu and Fe emission behavior, to consistently describe material interaction during welding under controlled conditions.

Despite these advances, the optical collection geometries in these studies were not implemented in a coaxial configuration with the welding beam, which limits signal robustness and scalability, particularly under high-power and high-speed laser welding conditions. Furthermore, a unified approach that combines process-generated plasma, a fully coaxial optical configuration, and LIBS-based spectrochemical interpretation without the use of an additional excitation laser remains lacking. While previous studies have demonstrated that LIBS and plasma spectroscopy can provide valuable information for welding quality monitoring, the realization of a fully coaxial spectroscopic monitoring configuration capable of robust operation under practical high-power laser welding conditions remains insufficiently explored.

In this study, we investigate a real-time, coaxial spectroscopic monitoring approach for assessing laser welding quality by utilizing plasma generated during continuous-wave (CW) laser welding. Unlike conventional LIBS systems that employ a dedicated pulsed laser for ablation, the plasma in this study is generated directly by the interaction between the high-power welding laser and the material. As a result, the collected emission originates from a laser-induced plasma, and the present approach follows the fundamental principle of LIBS while differing in its process-integrated implementation. In particular, the proposed method integrates a fully coaxial optical configuration, process-generated plasma, and LIBS-based spectrochemical analysis within a single-laser framework, eliminating the need for an additional excitation source. The proposed approach does not require an additional laser source for LIBS excitation and is designed to operate directly with the welding process. Copper–steel lap-joint welding was conducted using a high-power CW laser, while LIBS emission signals were collected

in a coaxial configuration to enable in situ monitoring of weld quality. The applicability of the method was examined under different welding conditions by analyzing characteristic Cu and Fe emission features, and the spectroscopic indicators were validated through weld cross-sectional analysis and mechanical shear load tests. The results demonstrate that the proposed coaxial LIBS-based monitoring scheme can effectively reflect variations in weld quality under practical high-power laser welding conditions.

2. Experimental setup

2.1. Materials and experimental setup for laser welding

C1100P copper and SPCD steel specimens, each with dimensions of $100 \times 30 \times 0.5 \text{ mm}^3$, were used for laser welding, and their chemical compositions are summarized in Table 1.

Laser welding experiments were performed using a high-power continuous-wave (CW) green laser (TruDisk 2021, TRUMPF) with a wavelength of 515 nm and a maximum output power of 2 kW. The laser beam was delivered to the welding head through a 20 m optical fiber with a core diameter of 100 μm . The output laser power was varied between 1.2, 1.4, and 1.6 kW during the experiments. The delivered beam passed through a galvanometer-based scanning head (intelliSCAN30, SCANLAB) and was focused onto the specimen surface using an f-theta lens with a focal length of 255 mm. After passing through the f-theta lens, the laser beam formed a spot diameter of approximately 182 μm at the focal plane. Linear scanning was applied along the welding direction, with scan speeds set to 250, 300, and 350 mm/s. Based on the combination of laser power and scan speed, the laser welded specimens were denoted as C1–C9, as summarized in Table 2. Copper and steel specimens were mounted on a dedicated welding jig in a lap-joint configuration. After overlapping the two sheets, laser welding was performed along a straight weld line with a length of 20 mm. A schematic illustration of the overall experimental setup is shown in Fig. 1.

2.2. Laser-induced breakdown spectroscopy (LIBS)

For spectroscopic measurements, an echelle spectrometer (Mechelle ME5000, Andor) equipped with an intensified CCD camera (iStar DH334T-18F-03, Andor) was employed to acquire emission spectra from laser-induced plasma generated during laser welding. The spectrometer covers a broad wavelength range from 200 to 975 nm with a spectral resolution of approximately 0.1 nm, enabling simultaneous detection of multiple elemental emission lines relevant to copper–steel welding. The ICCD gating parameters were fixed at a gate delay of 0 ms and a gate width of 80 ms for all measurements, in order to capture the intense plasma emission generated during the welding process while maintaining stable signal acquisition. To synchronize spectroscopic acquisition with the laser welding process, a delay generator was used to control the timing between the welding laser and the ICCD camera. The welding laser emission was detected by a photodetector, and the detected signal was used as a trigger to initiate the spectroscopic measurement. The temporal synchronization between the laser emission, ICCD gate opening, and data acquisition was verified using an oscilloscope, ensuring accurate time alignment during LIBS measurements.

Plasma emission was collected through an optical path configured coaxially with the welding laser beam to ensure consistent signal collection during high-speed beam scanning. The collected light passed

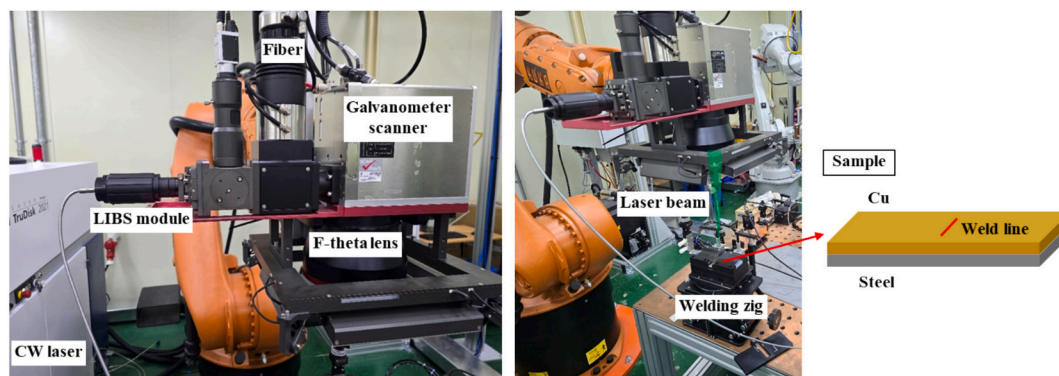
Table 1
Chemical composition of Cu and steel.

C1100P Cu wt%	Cu 99.983	Fe 0.0043	Sn 0.0010	Ni 0.0008	Pb 0.0005
SPCD steel wt%	Fe 99.8871	Mn 0.0990	S 0.0038	C 0.0020	P 0.0109

Table 2

Laser welding conditions used for each specimen.

Specimen	C1	C2	C3	C4	C5	C6	C7	C8	C9
Power [kW]	1.2	1.2	1.2	1.4	1.4	1.4	1.6	1.6	1.6
Scan speed [mm/s]	250	300	350	250	300	350	250	300	350

**Fig. 1.** Experimental setup for laser welding and LIBS measurement.

through the same f-theta lens and galvanometer scanner used for laser welding and was redirected toward the detection optics using a dichroic mirror. To suppress the intense laser radiation and to protect the ICCD detector from potential damage, a 514 nm notch filter was inserted into the collection path. The filtered plasma emission was then focused by a plano-convex lens (1-in. diameter, focal length 75 mm) into an optical fiber with a core diameter of 90 μm and a length of 2 m, which was coupled to the spectrometer via an SMA connector.

For each welding condition, ten LIBS spectra were acquired under identical optical and acquisition conditions. Repeated measurements were used to obtain statistically representative spectral data for subsequent analysis. This coaxial optical configuration enabled real-time acquisition of plasma emission spectra directly from the weld zone under varying welding conditions, providing robust spectral data for subsequent analysis of elemental emission behavior and weld quality.

2.3. Cross-sectional weld morphology

After laser welding, selected specimens were sectioned perpendicular to the welding direction. The samples were then mounted and mechanically polished using silicon carbide (SiC) abrasive papers, followed by fine polishing. The weld cross-sections were examined using an optical microscope (OM, HIROX) to evaluate the weld morphology and penetration depth under different welding conditions.

2.4. Shear load testing

Mechanical shear load tests were conducted using a universal testing machine (E43, MTS Systems) with a maximum load capacity of 50 kN. For each welding condition, three specimens were tested to ensure reproducibility. The tests were performed at a constant crosshead speed of 1 mm/min, and the maximum shear load at failure was recorded for each specimen.

3. Results and discussion

3.1. Laser welding quality assessment based on LIBS

In this section, the characteristics of the LIBS spectra acquired during laser welding are analyzed, and their relationship with welding conditions and weld quality is discussed. Prior to detailed spectral interpretation, preprocessing steps were applied to ensure reliable comparison

of emission features obtained under different welding conditions. These steps were designed to exclude non-informative spectral regions and to reduce baseline variations originating from the optical system and plasma background emission.

Although the spectrometer is specified to cover a wavelength range from 200 to 975 nm, the effective detection range in the present coaxial configuration was limited by transmission through the galvanometer scanner and the f-theta lens. As a result, wavelengths below approximately 390 nm and above 850 nm did not reach the detector, and no measurable signal was observed in these regions (Fig. S1(a)). For spectral preprocessing, data outside the analysis band ($\lambda < 300$ nm and $\lambda > 850$ nm) were therefore excluded from further analysis (Fig. S1(b)). Subsequently, constant baseline subtraction was performed to compensate for background emission and system-induced offsets. The baseline level was defined as the median intensity within the 300–350 nm range, where no prominent emission lines were observed, and this value was subtracted from the entire spectrum (Fig. S1(c)). In addition, due to the wavelength-dependent transmission characteristics of the optical components, no explicit spectral response correction was applied. Therefore, the analysis focuses on relative spectral features obtained under identical experimental conditions.

Fig. 2 compares representative LIBS spectra obtained from a Cu bead-on-plate specimen (Cu) and during laser welding of Cu–steel joints (Welding). The spectra are presented in two wavelength regions corresponding to (a) Cu emission lines and (b) Fe emission lines. The analysis focuses on identifying characteristic emission features relevant to welding quality assessment.

Under the Cu bead-on-plate condition, the spectra are dominated by strong Cu I and Cu II emission lines, including Cu II lines at 427.60 nm, 437.83 nm, and 450.59 nm, and Cu I lines at 458.70 nm, 465.11 nm, 467.47 nm, 470.46 nm, 793.31 nm, and 809.26 nm. The commonly used representative Cu I emission line at 521.82 nm was not observed in the present measurements because this wavelength lies within the attenuation band of the notch filter employed to suppress intense laser radiation. The detected emission lines reflect plasma emission originating from copper without interaction with the steel substrate and provide reference spectral characteristics under the applied optical configuration.

In contrast, the spectra acquired during laser welding exhibit additional emission features in the wavelength region shown in **Fig. 2(b)**. Several Fe I emission lines, including those near 396.31 nm, 396.93 nm, 397.77 nm, 398.40 nm, 404.58 nm, 428.99 nm, 430.79 nm, 432.58 nm,

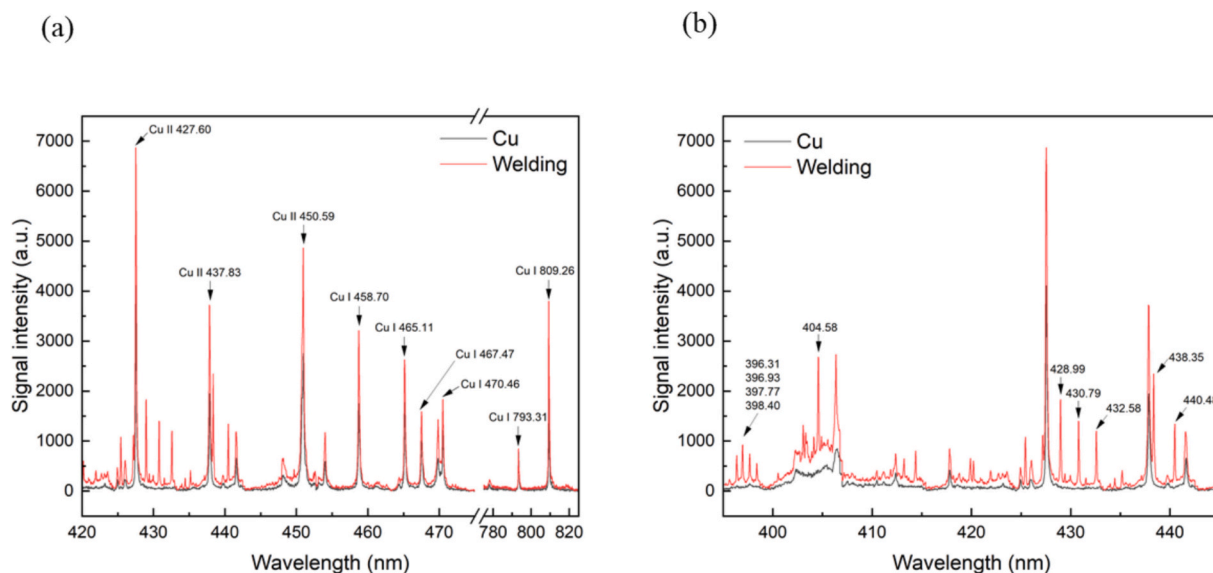


Fig. 2. Representative LIBS spectra acquired from (Cu) a Cu bead-on-plate specimen and (Welding) during laser welding of Cu–steel joints: (a) Cu emission lines and (b) Fe emission lines.

438.35 nm, and 440.48 nm, are observed exclusively under welding conditions. The appearance of these Fe emission lines indicates the involvement of the steel substrate in plasma generation during the welding process.

The presence of Fe-related emission lines during welding provides spectroscopic evidence of material interaction at the Cu–steel interface. Accordingly, the detection of Fe emission features can be used as an indicator for assessing welding quality, particularly in terms of

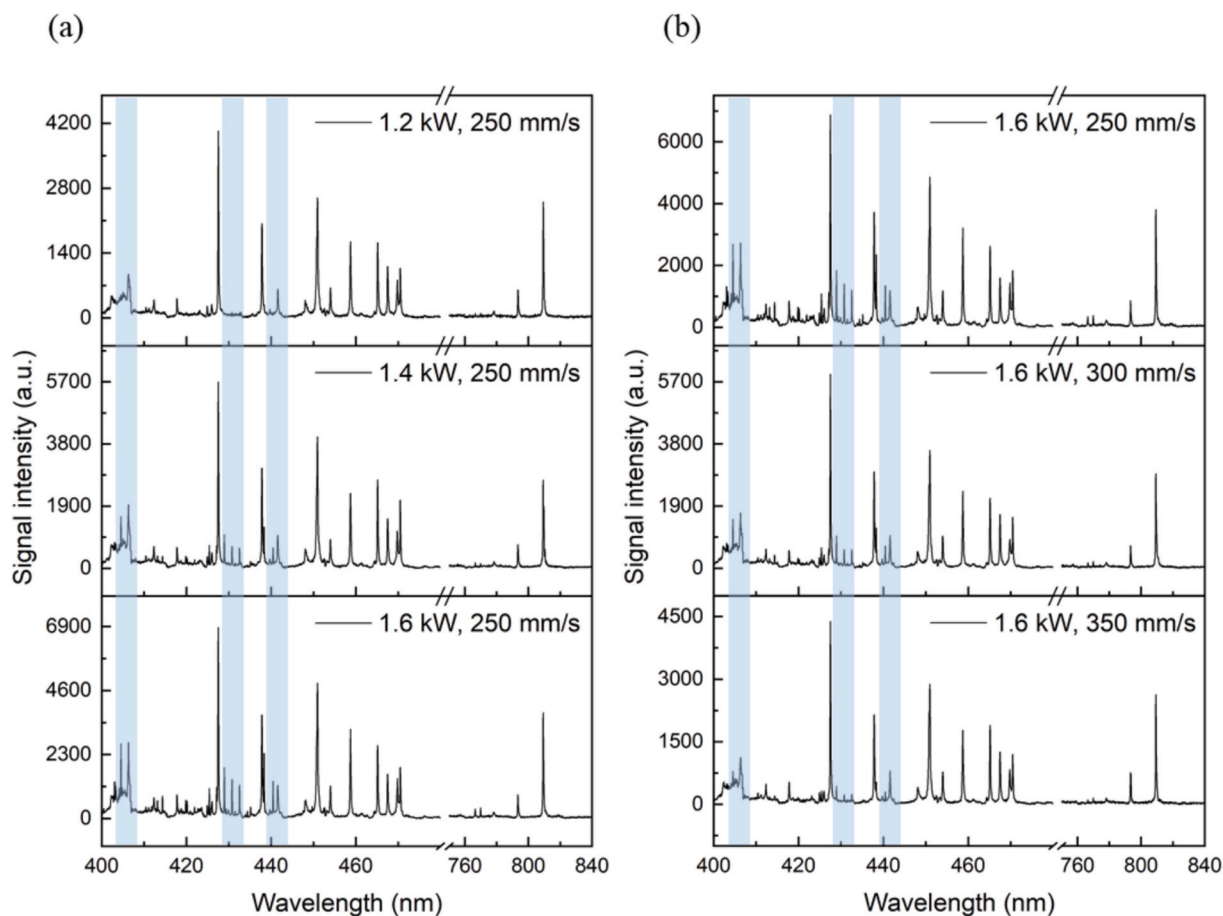


Fig. 3. LIBS spectra acquired during Cu–steel laser welding under different welding conditions: (a) variation of laser power at a fixed scanning speed of 250 mm/s and (b) variation of scanning speed at a fixed laser power of 1.6 kW. Blue-shaded regions indicate wavelength ranges corresponding to Fe emission lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

penetration and interfacial bonding.

Fig. 3 presents LIBS spectra acquired during Cu–steel laser welding under different laser powers and scanning speeds. The blue-shaded regions indicate wavelength ranges where Fe emission lines are observed. Fig. 3(a) shows the effect of laser power variation at a fixed scanning speed of 250 mm/s, while Fig. 3(b) illustrates the effect of scanning speed variation at a fixed laser power of 1.6 kW.

As shown in Fig. 3(a), an increase in laser power from 1.2 to 1.6 kW leads to a noticeable enhancement of Fe-related emission features within the highlighted regions. At lower laser power (1.2 kW), Fe emission lines are relatively weak, whereas higher laser power results in more pronounced Fe peaks. This trend suggests increased interaction between the molten copper and the underlying steel substrate as the laser power increases, reflecting deeper penetration and stronger material involvement during welding.

Fig. 3(b) shows the LIBS spectra acquired at different scanning speeds under a constant laser power of 1.6 kW. As the scanning speed increases from 250 to 350 mm/s, the intensity and prominence of Fe emission lines in the blue-shaded regions decrease. This behavior indicates a reduction in the contribution of the steel substrate to the plasma emission at higher scanning speeds, which can be attributed to decreased heat input and shallower penetration during the welding process.

The observed dependence of Fe emission features on laser power and scanning speed demonstrates that LIBS spectra sensitively reflect changes in welding conditions. In particular, the appearance and relative strength of Fe emission lines provide direct insight into the degree of interaction at the Cu–steel interface, highlighting the potential of in-situ LIBS monitoring for assessing welding quality during laser welding.

Fig. 4 presents the intensities of selected Cu neutral (Cu I) and singly ionized (Cu II) emission lines obtained from (a) bead-on-plate experiments and (b) Cu–steel laser welding under different laser powers and scanning speeds. For each experimental condition, the emission intensities were obtained by averaging ten repeated measurements.

In the bead-on-plate experiments (Fig. 4(a)), the highest Cu emission intensities are observed under condition C1 (1.2 kW, 250 mm/s). This condition corresponds to the lowest laser power combined with the slowest scanning speed, resulting in sufficient heat input per unit length while maintaining a relatively stable plasma generation regime. Under these conditions, Cu species are effectively excited and discrete Cu emission lines are clearly observed. For conditions with higher laser power than C1 at the same scanning speed, the plasma becomes more

intensely excited. In this regime, increased plasma density and optical thickness enhance self-absorption effects, which can limit the escape of emitted photons from the plasma volume. As a result, higher laser power does not necessarily lead to stronger observable Cu emission lines in bead-on-plate conditions.

In contrast, during Cu–steel laser welding (Fig. 4(b)), the Cu emission intensities increase with increasing laser power and decreasing scanning speed. Unlike the bead-on-plate condition, plasma generation during welding is strongly influenced by keyhole formation, melt pool dynamics, and interaction with the underlying steel substrate. Higher laser power and lower scanning speed promote deeper penetration and sustained plasma–material interaction, leading to enhanced excitation of Cu species and increased emission intensity.

These results demonstrate that the behavior of Cu emission lines differs fundamentally between bead-on-plate and welding conditions. While Cu emission in bead-on-plate experiments is governed primarily by plasma stability and optical thickness effects, Cu emission during welding reflects welding-specific plasma generation mechanisms associated with penetration and interfacial interaction. However, because Cu emission originates mainly from the copper material itself, variations in Cu emission intensity alone do not provide sufficient information to unambiguously assess welding quality. This distinction highlights the importance of considering process-dependent plasma characteristics and incorporating additional elemental information, such as Fe emission from the steel substrate, when interpreting LIBS signals for laser welding quality assessment.

As shown in Fig. 5, Fe emission intensities remain relatively low under conditions C1–C3, where limited interaction with the steel substrate is expected. A pronounced increase in Fe emission intensity is observed at condition C4, indicating the onset of significant penetration into the steel substrate. At conditions C5 and C6, the Fe emission intensities decrease again. These conditions correspond to increased scanning speeds, which reduce the effective heat input per unit length and limit the extent of steel involvement in the molten pool and plasma generation. As a result, the contribution of Fe species to the plasma emission is reduced, leading to lower Fe signal intensities. A sharp increase in Fe emission is observed at condition C7, where higher laser power combined with sufficient heat input promotes deeper penetration and more active interaction with the steel substrate. At conditions C8 and C9, the Fe emission intensities decrease compared to C7, which can be attributed to further increases in scanning speed and changes in welding dynamics that reduce sustained interaction with the steel.

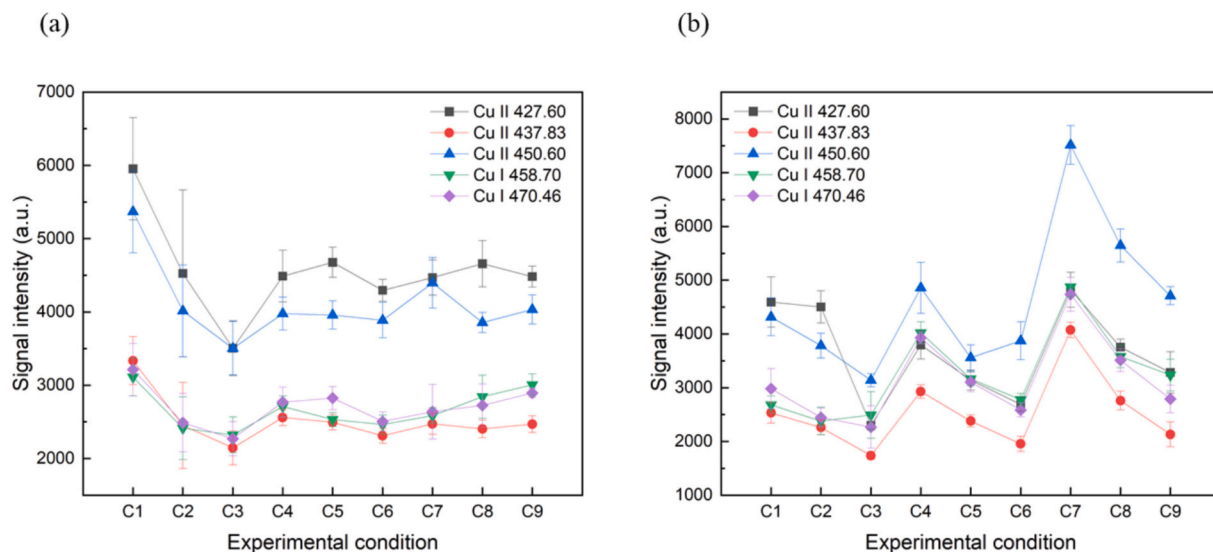


Fig. 4. Intensities of selected Cu I (neutral) and Cu II (singly ionized) emission lines obtained from (a) bead-on-plate experiments and (b) Cu–steel laser welding under different laser powers and scanning speeds.

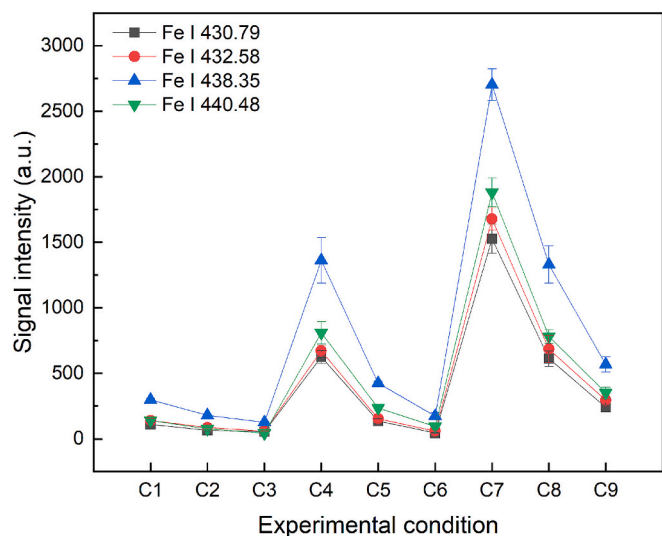


Fig. 5. Intensities of selected Fe I emission lines acquired during Cu–steel laser welding under different experimental conditions.

Overall, Fe-related LIBS signals provide more direct information on penetration behavior and interfacial interaction than Cu emission alone, as Fe emission originates exclusively from the steel substrate. However, despite this strong dependence on welding conditions, Fe emission intensity does not provide a straightforward quantitative criterion for welding quality assessment. The absence of a clear threshold or reference value makes it difficult to define welding quality based solely on Fe signal intensity, indicating that Fe-based LIBS indicators are most effective when used in a relative or comparative manner across different welding conditions.

Fig. 6 shows the Cu/Fe intensity ratios obtained from LIBS spectra acquired during Cu–steel laser welding under different experimental conditions. The Cu/Fe ratio was calculated by dividing the intensities of multiple Cu emission lines by the intensity of a representative Fe I emission line at 438.35 nm. This ratio reflects the relative contribution of copper and steel species to the laser-induced plasma during welding.

Physically, Fe emission originates exclusively from the steel substrate and appears only when the molten pool and plasma formation region involve the steel side of the joint. An increase in Fe intensity

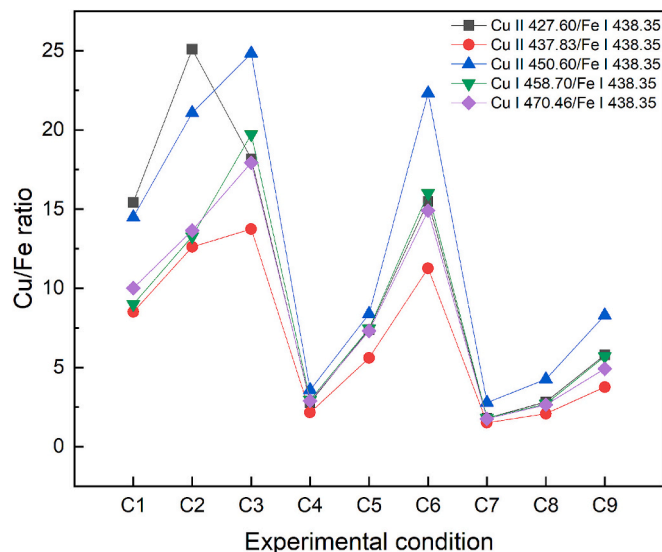


Fig. 6. Cu/Fe intensity ratios calculated from LIBS spectra acquired during Cu–steel laser welding under different experimental conditions.

therefore indicates enhanced steel participation, which is closely associated with penetration and interfacial interaction. In contrast, Cu emission is present under both bead-on-plate and welding conditions and can be influenced by plasma regime, optical thickness, and keyhole dynamics. By comparing Cu and Fe emission intensities through their ratio, variations in overall plasma emission level are mitigated, and changes related to material interaction at the Cu–steel interface are emphasized.

A low Cu/Fe ratio reflects an increased contribution of Fe emission relative to Cu emission, implying substantial involvement of the steel substrate in plasma generation. In the present results, conditions C4, C7, and C8 exhibit Cu/Fe values below 5 for all Cu emission lines considered. The consistent reduction of the Cu/Fe ratio across different Cu lines suggests that these conditions correspond to a common welding regime characterized by strong steel participation rather than being dominated by copper surface emission.

Mechanistically, such low Cu/Fe ratios can be interpreted as signatures of effective penetration and interfacial bonding. Under these conditions, the keyhole and molten pool extend sufficiently into the steel substrate, allowing Fe species to enter the plasma formation zone and contribute significantly to the observed emission. This behavior is consistent with the requirements for Cu–steel joining, where adequate interaction with the steel side is necessary to form a metallurgically bonded interface instead of a superficial Cu melt layer.

Although Fe-related signals more directly reflect penetration behavior than Cu emission alone, the Cu/Fe ratio does not provide a universal quantitative threshold for welding quality. The absolute value of the ratio depends on the experimental setup, including optical collection efficiency and spectral response. Therefore, the Cu/Fe ratio is most appropriately used as a relative indicator within a given system. Nevertheless, the observation that Cu/Fe values below 5 are consistently associated with welding conditions involving strong steel participation suggests that this ratio can serve as a practical criterion for monitoring welding quality under fixed experimental conditions.

3.2. Cross-sectional weld morphology

As shown in Fig. 7, representative cross-sectional images corresponding to selected welding conditions illustrate clear differences in penetration behavior and weld morphology. Condition C7 (1.6 kW, 250 mm/s) exhibits sufficient penetration into the steel substrate, indicating strong interaction between the molten copper and the steel. This condition is consistent with the high Fe emission intensity observed in Fig. 5 and the relatively low Cu/Fe ratio shown in Fig. 6, representing a stable welding regime with effective interfacial mixing.

At condition C5 (1.4 kW, 300 mm/s), partial penetration into the steel substrate is observed. Although the weld morphology remains relatively stable, the reduced penetration depth compared to C7 indicates decreased interaction with the steel substrate. This observation corresponds to moderate Fe emission intensity and intermediate Cu/Fe ratio values, reflecting a transitional welding condition.

In contrast, condition C3 (1.2 kW, 350 mm/s) shows significantly reduced penetration, with the molten pool largely confined within the copper layer. This limited interaction with the steel substrate corresponds to weak Fe emission signals and relatively high Cu/Fe ratios, indicating insufficient welding conditions.

It should be noted that several dark, rounded features observed in the cross-sectional images are attributed to redeposited spatter or polishing artifacts rather than internal defects such as cracks or porosity. These features are considered unrelated to the penetration behavior discussed in this study.

Overall, the cross-sectional observations confirm that variations in Fe emission intensity and Cu/Fe ratio are closely correlated with changes in penetration depth and interfacial interaction during welding. Deeper penetration and stronger interfacial interaction are associated with improved joint formation and mechanical performance, indicating

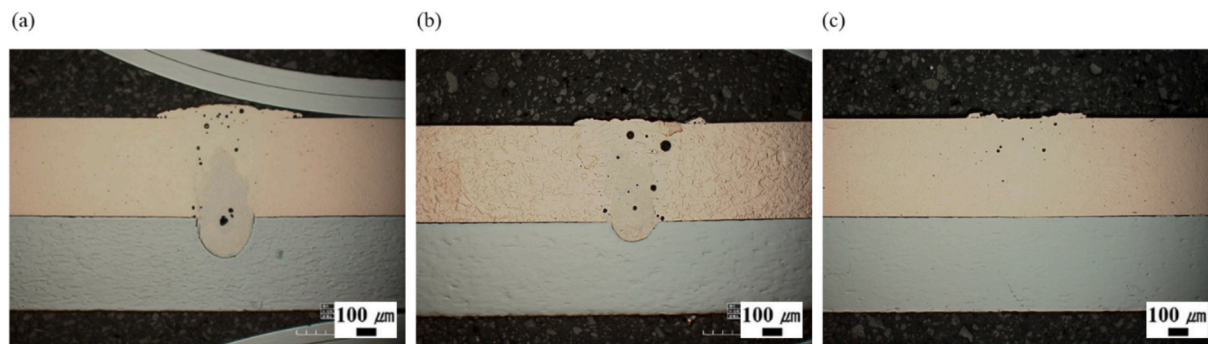


Fig. 7. Representative cross-sectional images of Cu-steel welds under selected welding conditions: (a) C7, (b) C5, and (c) C3.

a direct link to welding quality. These results support the interpretation that LIBS-based indicators provide reliable qualitative information on weld quality when used in a comparative manner across different welding conditions.

3.3. Mechanical shear load evaluation

Shear load represents the maximum force that a welded joint can withstand before failure under shear deformation and is a direct measure of the mechanical integrity of the joint. For dissimilar metal joints such as Cu-steel welds, shear load is particularly relevant because it reflects the effectiveness of metallurgical bonding at the interface rather than simple surface adhesion or localized melting.

A high shear load indicates that the applied load is effectively transferred across the Cu-steel interface, implying sufficient penetration into the steel substrate and the formation of a mechanically robust, metallurgically bonded joint. In contrast, low shear load values suggest inadequate interfacial bonding, where the joint strength is governed by weak mechanical interlocking or superficial copper melting without meaningful interaction with the steel substrate.

As shown in Fig. 8, the variation in shear load across different experimental conditions closely follows the trends observed in the Cu/Fe ratios derived from LIBS measurements (Fig. 6). Conditions exhibiting low Cu/Fe ratios, particularly C4, C7, and C8, correspond to significantly higher shear loads. This correspondence indicates that increased steel participation during welding, as reflected by enhanced Fe emission relative to Cu emission, leads to improved penetration and

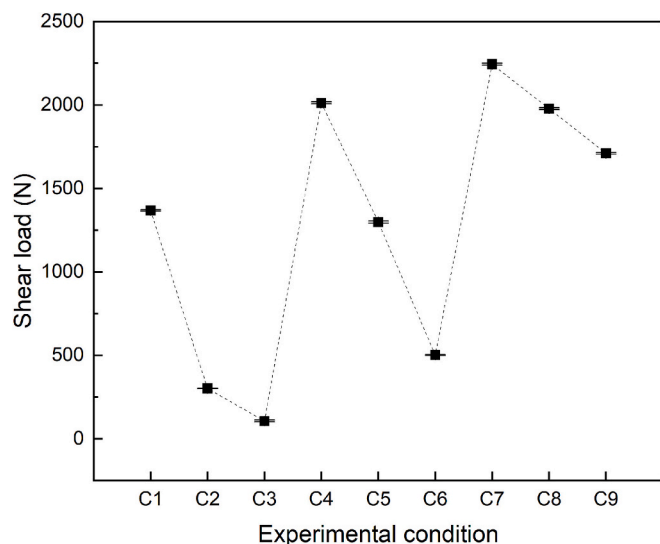


Fig. 8. Shear load of Cu-steel welded joints measured under different experimental conditions.

stronger interfacial bonding, which directly translates into higher mechanical strength.

Conversely, conditions with high Cu/Fe ratios show reduced shear loads, confirming that Cu-dominated plasma emission is associated with insufficient steel involvement and poor joint strength. This clear relationship demonstrates that shear load serves as a definitive post-process validation metric for welding quality, while the LIBS-derived Cu/Fe ratio provides a process-integrated indicator that captures the same underlying bonding mechanisms during welding.

3.4. Investigation of spectral peak shift

Fig. 9 presents time-resolved LIBS spectra of the Cu II emission line at 427.60 nm acquired under bead-on-plate and welding conditions at a laser power of 1.6 kW and a scanning speed of 250 mm/s. At zero gate delay (Fig. 9(a)), a clear peak shift is observed under welding conditions compared to the bead-on-plate case. Similar tendencies were also detected for other Cu emission lines; however, the Cu II 427.60 nm line is shown here as a representative example due to its relatively high intensity and good spectral isolation, which allow reliable observation of peak position changes.

To determine whether the observed peak shift originates from intrinsic material effects, such as stress or lattice distortion, or from plasma-induced effects, time-resolved measurements were conducted with gate delays of 30, 50, and 70 ms (Figs. 9(b)–(d)). When a gate delay is applied, no discernible peak shift is observed for either condition. Although the overall emission intensity decreases with increasing delay due to plasma cooling, the peak position remains unchanged.

The absence of peak shift at delayed gate times indicates that the shift observed at zero delay is not associated with permanent or stress-related material effects. Instead, it suggests that the shift is governed by transient plasma conditions present immediately after plasma formation. Plasma-induced spectral effects, such as Stark shift and line asymmetry, are known to be pronounced under high electron density conditions and to disappear rapidly as the plasma decays. The observed time-resolved behavior is fully consistent with this plasma-dominated mechanism.

The occurrence of the peak shift exclusively at zero delay under welding conditions can be attributed to the distinct plasma environment generated during laser welding. Compared to bead-on-plate experiments, laser welding involves keyhole formation, deeper molten pools, and intense metal vapor generation from both Cu and steel, resulting in higher and more transient electron densities in the laser-induced plasma. These conditions favor the appearance of plasma-induced spectral shifts immediately after plasma generation, while such effects are absent once the plasma cools.

It should also be noted that, even at delayed gate times where no peak shift is observed, the emission intensity under welding conditions remains higher than that under bead-on-plate conditions. This indicates that the disappearance of the peak shift is linked to reduced electron

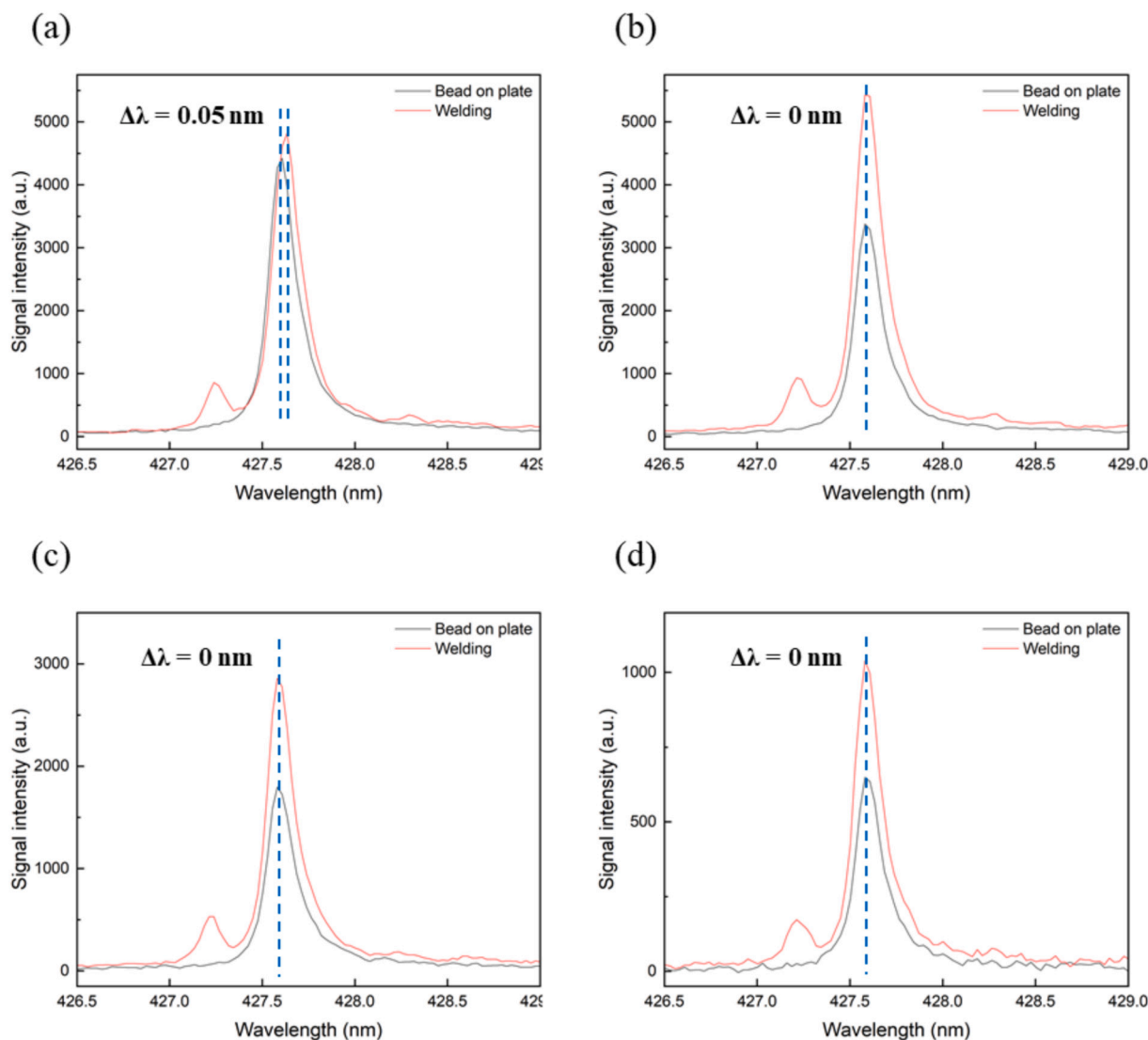


Fig. 9. Time-resolved LIBS spectra of the Cu II 427.60 nm line acquired under bead-on-plate and welding conditions at gate delays of (a) 0 ms, (b) 30 ms, (c) 50 ms, and (d) 70 ms.

density, whereas the emission intensity at delayed times is primarily governed by the number of excited emitting species. During laser welding, sustained metal vapor generation and a larger plasma volume lead to a higher population of excited Cu species, allowing stronger emission to persist after plasma cooling without accompanying peak shifts.

Accordingly, the LIBS results discussed in the preceding sections, which are based on emission intensities and intensity ratios of Cu and Fe, can be interpreted without significant interference from transient peak position variations under the present experimental conditions.

4. Conclusions

In this study, laser-induced breakdown spectroscopy (LIBS) was applied for in-situ monitoring of Cu-steel laser welding using a single continuous-wave welding laser and a coaxial optical configuration. Emission spectra generated during welding were successfully acquired in real time without an additional excitation source, demonstrating the feasibility of process-integrated LIBS monitoring.

Analysis of Cu and Fe emission behavior showed that Cu emission alone cannot unambiguously indicate welding quality, as Cu-related signals are present under both bead-on-plate and welding conditions

and are strongly affected by plasma characteristics. In contrast, Fe emission originates exclusively from the steel substrate and appears only when effective penetration and interfacial interaction occur, providing more direct information on welding behavior. However, the absolute intensity of Fe emission does not provide a clear quantitative criterion for welding quality.

To overcome this limitation, the Cu/Fe intensity ratio was introduced as a relative indicator reflecting material interaction during welding. Welding conditions exhibiting Cu/Fe ratios below approximately 5 were consistently associated with strong Fe participation and significantly higher shear loads, indicating improved interfacial bonding and mechanical joint strength. In addition, cross-sectional analysis of the welds confirmed that these conditions correspond to increased penetration depth and enhanced interaction with the steel substrate. Although the Cu/Fe ratio does not represent a universal quantitative threshold, it provides a practical and process-relevant criterion for distinguishing welding regimes under fixed experimental conditions.

Time-resolved LIBS measurements confirmed that spectral peak shifts observed under welding conditions are transient plasma-related effects occurring immediately after plasma formation and do not persist at delayed acquisition times. This observation supports the interpretation of the LIBS results primarily based on emission intensities

and intensity ratios without significant interference from peak position variations.

Overall, this work demonstrates that LIBS-based monitoring using Cu and Fe emission behavior, particularly through the Cu/Fe intensity ratio, provides a robust and physically meaningful approach for assessing Cu-steel laser welding quality. The proposed coaxial, single-laser LIBS configuration offers a promising route for real-time process monitoring and quality evaluation in dissimilar metal laser welding. By directly collecting emission from the laser-material interaction zone, the coaxial configuration enables spectroscopic signals to be closely linked to the ongoing welding process. While the use of a dedicated LIBS laser can allow spatially selective probing, the process-integrated approach adopted in this study provides advantages in terms of simplicity, robustness, and practical applicability to high-power laser welding environments.

CRedit authorship contribution statement

Hyun Jong Yoo: Writing – original draft, Investigation. **Heeshin Kang:** Supervision. **Hyeongwon Kim:** Writing – original draft, Investigation. **Geon Lim:** Investigation, Data curation. **Soojin Choi:** Writing – review & editing, Investigation.

Author statement

All authors have contributed to this work and approved the final version of the manuscript.

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The authors declare that they have no competing financial interests.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.sab.2026.107541>.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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